

HILLIARD SMART

CHRISTMAS LIGHTING

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HOW DID WE GET HERE?

- Smart Christmas Lighting for Hilliard is not a business idea
- We initially planned on building street lamps to prevent light pollution
- We found issues in the research process, and abandoned the project
- We were told about a separate project related to lighting
- What is “reactive lighting”



Our original Idea

HOW DOES IT WORK?

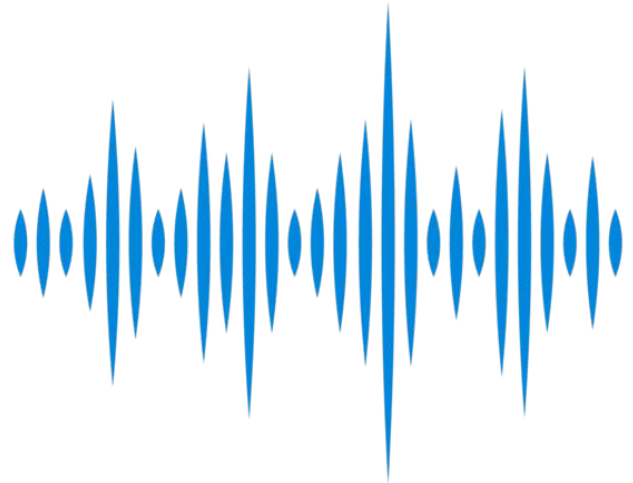
- The audio is collected and processed, sending signals in sync with the music
- The signals are sent to a central server
- The server directs the signals to a controllable user interface that determines specific instructions for the lights
- The system sends the light instructions to lighting boxes
- We control six smaller RGB lighting trees, and 1 large regular lighting tree



AUDIO INPUT & PROCESSING

The system to process the audio so that the individual lighting units react to the music

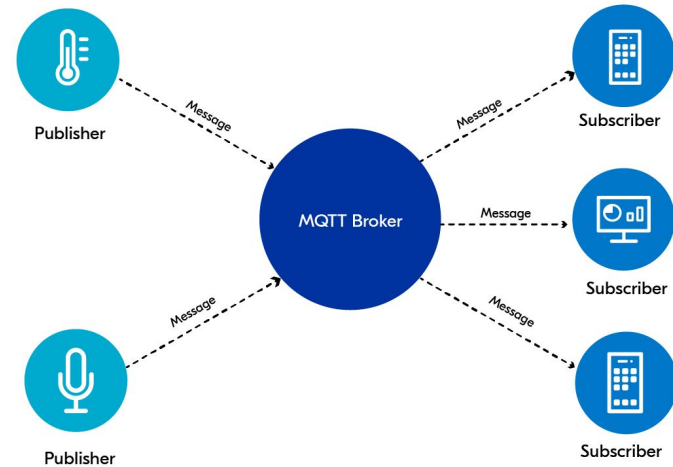
- Audio is played on a laptop while code is running in the background listening
- The code takes a chunk of audio and separates it into frequency ranges to use like bass, mid, and treble
- Those frequency ranges are put through filters to isolate characteristics of the audio
- The code sends a triggers to the central server based on the music



CENTRAL SERVER

The system to manage data flow and sync light displays

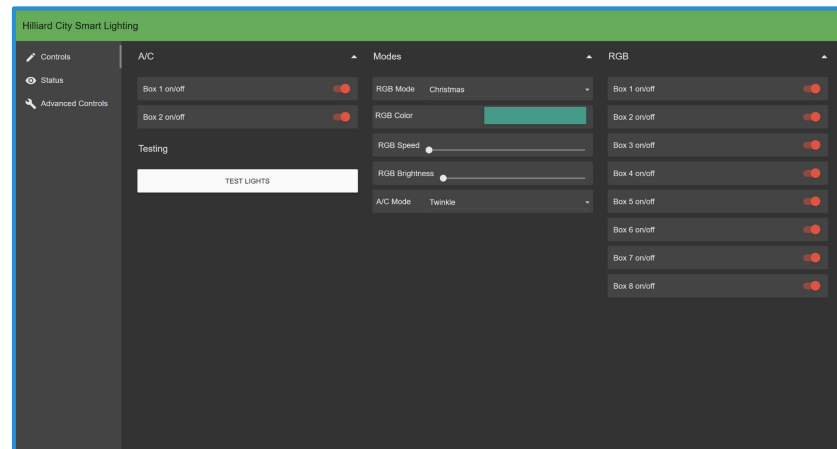
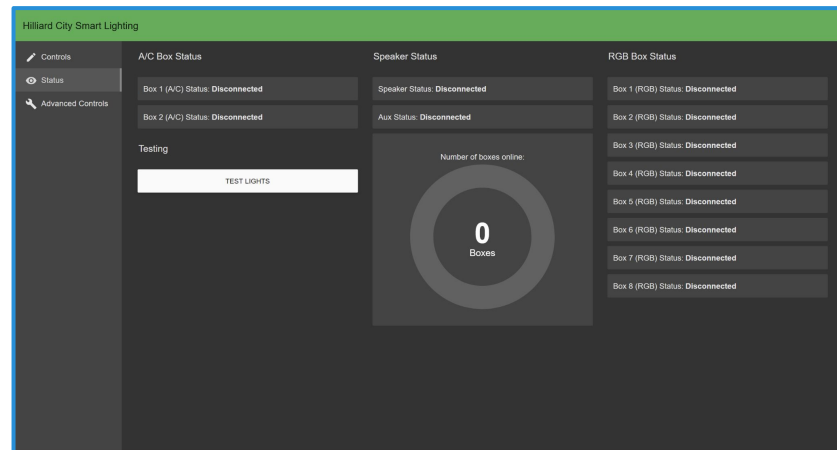
- The system runs on MQTT communication protocol (broker & clients)
- The broker is run on a Raspberry Pi
- Each box is assigned 2 unique topics, one for connection status and one lighting signals
- There are usernames and passwords for each part of the system



NODE-RED

An interactive user interface that controls and monitors data flow through MQTT

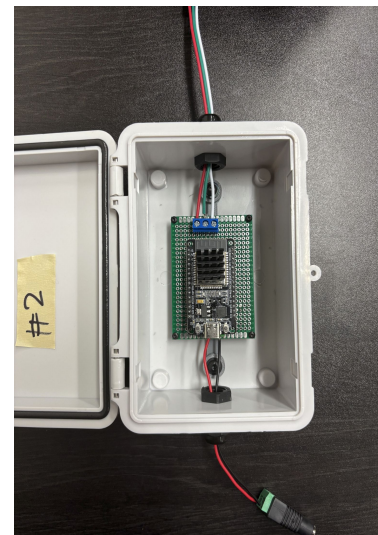
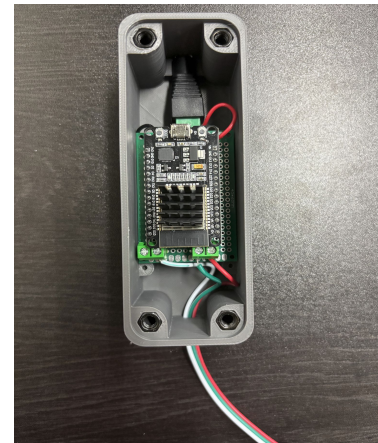
- Receives the trigger from the MQTT broker
- Sends out a signal corresponding to each box's respective mode
- Shows the connection status of each box
- Secured with usernames and passwords
 - One login for admins, one for users



RGB LIGHTING UNIT

The device controlling lights at each tree

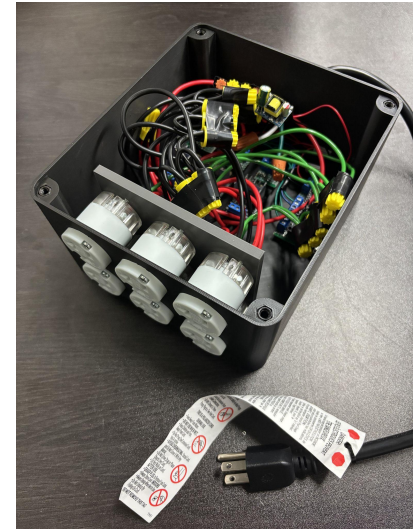
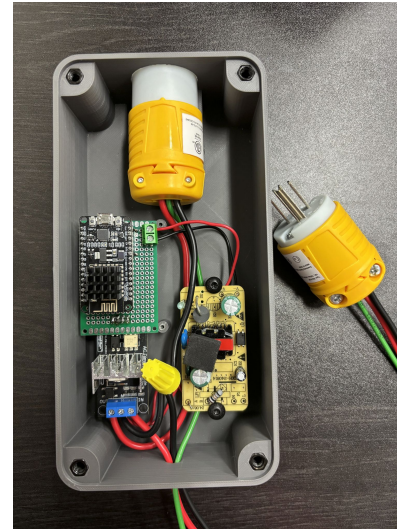
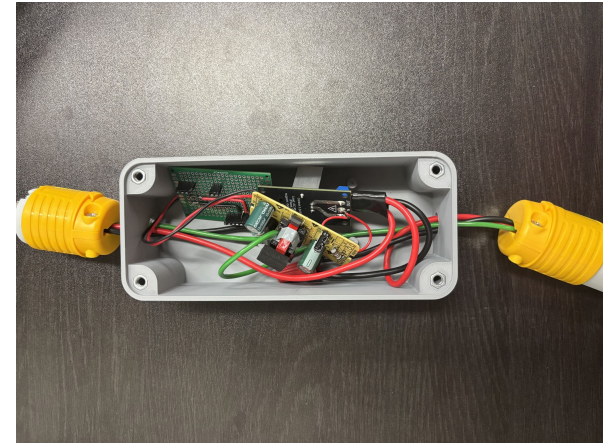
- What is RGB?
- Uses a microcontroller (ESP32) to connect to the server
- Currently designed to work with 200 leds (66 feet) of RGB leds
- Weatherproof design
- We have made 8 of them to be implemented into Old Hilliard



REGULAR LIGHTING BOXES

The device controlling lights at the big christmas tree

- Uses a microcontroller (ESP8266) to connect to the server
- Uses a pulse width module to brighten and dim the lights
- Designed for the big christmas tree so it features 6 outlets for the tree



FUTURE PLANS

- We will continue working until we finish development of the system
 - Decreasing delay
 - Better user interface for music processing
 - More power for longer RGB strings
- We will set up the system in Hilliard when Christmas is coming up
 - Connecting the system to Wifi
 - Configuring the system to the existing lights
 - Setting up a computer to run the music processing



THANK YOU!

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